Azrul Amaline

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Career Summary

Software Engineer and Computer Science and Engineering graduate with hands-on professional experience in **cross-platform mobile development, backend engineering, and AI-powered applications**. Experienced in designing scalable Flutter applications, building RESTful APIs, implementing secure authentication, integrating third-party services, and developing maintainable software architectures. Proficient in **Flutter/Dart, Python, Django, FastAPI, Firebase, PostgreSQL, and SQL**, with experience in state management, real-time communication, and multi-tenant backend systems. Hands-on exposure to **Linux, Docker, GitHub Actions CI/CD, Nginx, and cloud/VPS deployment**, alongside embedded systems and Raspberry Pi integration. Conducted deep learning research on distributed-system log anomaly detection using **GRU and Multi-Head Attention**, with ongoing interests in continual learning, intelligent automation, DevOps, and cloud-native software engineering.

Experience

Junior Software Developer Intern (Flutter) Nov 2025 – Feb 2026

MemoNEET – Remote, Bangladesh

- Implemented **scheduled notifications, learning-task reminders, and quiz-based workflows** in a Flutter mobile application.
- Debugged Flutter user interfaces and collaborated on feature delivery using **Git and GitHub**.

Selected Engineering Projects

DevDroid – Android Code Editor & Developer Workspace [GitHub]

Flutter, Dart, Termux, SSH, Git

- Built an Android-first development workspace with a **multi-tab syntax-highlighted editor, project explorer, Termux-backed terminal**, and in-app Git workflows.

ResearchLens AI – Research Paper Analysis Platform [GitHub]

Next.js, FastAPI, PostgreSQL, FAISS, Docker

- Developed a **RAG-based research workspace** for academic PDF ingestion, semantic search, page-linked answers, and multi-paper comparison.

Bangladesh Job Preparation – Multi-Track Learning Platform [GitHub]

Flutter, BLoC, Firebase, FastAPI

- Designed a multi-track examination-preparation system using **six Firebase projects**, a FastAPI content gateway, centralized authentication, and reusable quiz flows.

Syncora – Real-Time Multi-Tenant Business API [GitHub]

Django, DRF, PostgreSQL, WebSockets

- Engineered a **tenant-isolated business backend** with JWT and role-based permissions, inventory and sales transactions, audit logs, and live notifications.

Schedula – Academic Management Platform [GitHub]

Flutter, Firebase, Generative AI

- Created an academic app for schedules, notes, assignments, announcements, student profiles, **real-time chat**, and AI-assisted learning.

Lecturer AI – Smart Lecture Study App [GitHub]

Flutter, Speech-to-Text, AI APIs, ML Kit

- Developed an AI study-app concept with **lecture transcription, summaries, quiz generation**, and OCR-assisted note digitization.

Research

Undergraduate Thesis – System Log Anomaly Detection Completed 2026

Patuakhali Science and Technology University

Log Anomaly Detection in Distributed Computing Environments Using an Optimized Recurrent Neural Network with Multi-Head Attention

- Developed a **GRU + Multi-Head Attention** sequence-classification pipeline and evaluated **HDFS, OpenStack, and BGL** log datasets using F1-score and ROC-AUC.

DynaLog – Continual Learning for Evolving System Logs Ongoing

Independent research; not published

- Investigating how to distinguish **concept drift from anomalies** in evolving log streams through temporal models and continual adaptation.

Technical Skills

Mobile Engineering: Flutter, Dart, BLoC, Riverpod, Firebase, REST API Integration
Backend & Web: Python, Django, FastAPI, Next.js, REST APIs
AI & Data: PyTorch, RAG, FAISS, PostgreSQL, MySQL
Languages: Dart, Python, JavaScript, Java, C++, SQL
DevOps & Tools: Linux, Git, GitHub, Docker, Postman; learning AWS, CI/CD, Terraform, Kubernetes

Education

Patuakhali Science and Technology University (PSTU) 2026
B.Sc. (Engineering) in Computer Science and Engineering

Maheshpur Govt. Degree College 2020
Higher Secondary Certificate (Science)

Maheshpur Govt. Pilot High School 2018
Secondary School Certificate (Science)

Achievements & Activities

- **Vice President, CSE Club, PSTU:** Supported technical events, programming contests, workshops, and student learning initiatives.
- **President, Greater Jashore Student Welfare Association, PSTU:** Led student welfare initiatives, events, and community engagement.
- **Competitive Programming:** Solved 385+ Codeforces problems.
- **National Cyber Drill 2022:** Ranked 32nd nationally.
- **Bdapps Mobile App Challenge:** Participated in a national-level app-development competition.

Certifications & Training

Python & Backend: Python (Django) Training – EDGE / BCC; Python Masterclass 2022 – Udemy
AI & Machine Learning: Python for Machine Learning; Python for Deep Learning – Udemy
Networking & Systems: Basics of Computer Networking – Great Learning; Introduction to Packet Tracer – Cisco Networking Academy; Concepts of Operating Systems – Udemy
Cybersecurity: National Cyber Drill 2022 – BGD e-GOV CIRT (N-CERT)
Languages: Bangla (Native), English (Professional Working), Hindi (Conversational)